

WILLIAM NIEMIEC

PH.D. STUDENT | SOFTWARE ENGINEERING IN HEALTHCARE

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William Niemiec has 5 years of experience as a full-stack developer and is a Ph.D student at the Institute of Informatics of the Federal University of Rio Grande do Sul (UFRGS) since 2024. He also works as a senior systems developer at Paytrack in Brazil. He has managed several projects, as well as developed various systems, including back-end, front-end, and mobile applications. In parallel, he has been researching software engineering in healthcare since 2021, attending health conferences, and organizing workshops with healthcare professionals. His studies include interoperability protocols such as HL7 and FHIR. He also has research interests in Software Engineering, Component-Based Software Engineering, Model-Driven Development, and Artificial Intelligence applied in healthcare.

FORMAL EDUCATION

Ph.D. in Computer Science <i>Federal University of Rio Grande do Sul</i> <i>A Methodology for Defining a No-Code Solution: Healthcare Domain Case Study</i>	2024 - Present
Bachelor of Computer Science <i>Federal University of Rio Grande do Sul</i> <i>Take: a no-code platform for the development of mHealth applications based on care plans</i>	Aug 2018 - Oct 2022

SOME PUBLISHED PAPERS

IEEE/ACM 13th International Conference on Mobile Software Engineering and Systems (MOBILESoft) <i>A Methodology for Generating Mobile Applications Through Large Language Models</i>	2026
IEEE International Symposium on Computer-Based Medical Systems (CBMS) <i>Leveraging Natural Language Processing for mHealth Development: A Component-Based Approach Using Nursing Taxonomies</i>	2025
Journal of Systems and Software (JSS) <i>Towards a component-based framework for mHealth apps: Bridging the gap between the nursing domain language and the computation domain</i>	2025
Brazilian Symposium on Computing Applied to Health (SBCAS) <i>Avaliando a qualidade de aplicativos de saúde através da metodologia uMARS e do uso de personas</i>	2025
Workshop on the Implications of Computing in Society (WICS) <i>Artificial intelligence discrimination: How to deal with it?</i>	2022
Brazilian Symposium on Software Engineering (SBES) <i>ExecutionFlow: a tool to compute test paths of Java methods and constructors</i>	2021

PEER REVIEWER WORKS

1st Regional School of Machine Learning and Artificial Intelligence of the Southern Region <i>Porto Alegre, Rio Grande do Sul, Brazil</i>	2025
Xth Workshop on Scientific Initiation in Information Technology and Human Language <i>Fortaleza, Ceará, Brazil</i>	2025

LANGUAGES

- English: Advanced
- Spanish: Intermediate
- Portuguese: Fluent

HONORS

International Young Scientist Awards <i>Issued by ScienceFather</i>	2025
Young Researcher Award <i>Issued by the Federal University of Rio Grande do Sul</i>	2022

ACADEMIC JOBS

Federal University of Rio Grande do Sul | Teaching Practice

Oct 2024 – Jun 2025

My responsibilities included teaching classes, managing a Moodle platform, and assisting the primary professor. This experience improved my ability to explain complex technical concepts clearly and effectively.

Federal University of Rio Grande do Sul | Undergraduate Research Assistant

Feb 2020 – Oct 2022

My early career included foundational roles in scientific research, focusing on full-stack development for a no-code health platform and work in data science and machine learning.

RESEARCH PROJECTS

REDI-SUS Intelligent Digital Health Network: Diagnosis, Care Plans, and Remote Monitoring

2025 – Present

It is proposed the creation of an Intelligent Digital Health Research Network aiming to develop a modular and interoperable platform that supports healthcare professionals and patients throughout all stages of the care journey, from diagnosis to long-term monitoring. The solution encompasses the development of medical devices, mobile applications, and artificial intelligence (AI) systems for diagnosis, as well as digital twins and customizable mHealth solutions focused on remote monitoring and individualized care management for patients with different clinical profiles, including chronic diseases, neglected diseases, and high-impact conditions such as heart transplantation. The research provides for interoperability with SUS Digital (HL7 FHIR) and validation of technologies in university hospitals, Basic Health Units, and home care services through partnerships with the RUTE Network and Telehealth programs. The project will implement data governance, including secure data sharing protocols in compliance with the LGPD, and scalable technological integration models that enable national expansion. The main focuses include the validation of medical devices, sensors, and AI solutions for early diagnosis and clinical decision support; the development of continuous remote monitoring and follow-up technologies; and support for personalized care plans that accompany the patient throughout the entire health journey. More assertive and early clinical decisions, a reduction in preventable hospitalizations, and increased treatment adherence are expected, strengthening a patient-centered, equitable, and sustainable care model aligned with the principles of the SUS.

No-code platform for mHealth applications

2023 – Present

The overall goal of this project is to identify a domain-specific language and specify basic building blocks for defining customizable healthcare plans, and to develop a no-code platform so that healthcare professionals can act as citizen developers and create applications to support medium/long-term health treatments. The platform will be built upon the essential elements of health treatments that can be interconnected to create individual care plans. The proposed platform is classified as no-code because it must provide a user-friendly interface in the language of the healthcare professional (HCP), without requiring any knowledge of programming logic to be used.

Automation of clinical instrument generation through prompt engineering

2025 – 2026

This research project aims to explore the feasibility and effectiveness of using Artificial Intelligence (AI) to automate the creation of digital clinical instruments on the Otus Solutions platform. Otus, a company born from the experience of the renowned Longitudinal Study of Adult Health (ELSA-Brasil), offers an integrated solution to simplify and optimize clinical research. Currently, the instantiation of instruments on its platform, 'Otus Studio', is a manual and time-consuming process. The methodology includes a literature review, the iterative development and testing of prompts, the comparison of AI models, and the encapsulation of the best-performing model into an API (Application Programming Interface). The project includes a two-phase validation process: an internal validation, conducted by the partner company's team, and an external validation, with healthcare professionals and clinical researchers. It is expected that the automation will significantly reduce the time required to create the instruments, optimizing researchers' workflow and reducing the technical barrier to the digitalization of clinical instruments.

Using mobile applications for health monitoring/management

2023 – 2025

Interaction with healthcare teams and volunteers to understand the use of mHealth in practice and identify essential elements of health treatments that can be translated into interconnectable computational elements for the creation of care plans.

ACADEMIC ADVISORY

Course Conclusion Paper Federal University of Rio Grande do Sul <i>Marcello Saretta Tomazi. Architecture for Dynamic Adaptation of Clinical Terminology via Large Language Models: A Case Study on the Takere Platform</i>	Mar 2026 - Present
Course Conclusion Paper Federal University of Rio Grande do Sul <i>Renan Kummer.. Mapping Remote Monitoring Data to HL7 FHIR Standard Resources: A Case Study on the Takere Platform</i>	Mar 2026 - Present
Course Conclusion Paper Federal University of Rio Grande do Sul <i>Lucca Franke Kroeff. Use of LLMs for Simplifying Technical Language in mHealth Platforms</i>	Mar 2026 - Jul 2026
Course Conclusion Paper Federal University of Rio Grande do Sul <i>Sofia de Moraes Sauter Braga. An exploratory study of the use of Claude in the development and evaluation of a mobile mHealth interface</i>	Mar 2026 - Jul 2026
Course Conclusion Paper Federal University of Rio Grande do Sul <i>Yasmin K. B. Zebrowski. Gamification Mechanisms as Motivational Factors for Adherence to Medical Treatments</i>	Mar 2025 - Dez 2025
Course Conclusion Paper Federal University of Rio Grande do Sul <i>Richard L. Ramos. Integration of Voice Assistants and Wearable Devices in Personalized Care Plans: An mHealth Approach to Support Patient Adherence</i>	Mar 2025 - Dez 2025
Course Conclusion Paper Federal University of Rio Grande do Sul <i>Marlize L. Batista. Evaluation of the use of AI agents in the validation activity of healthcare care plans</i>	Nov 2024 - Dez 2025
Course Conclusion Paper Federal University of Rio Grande do Sul <i>Leonardo V. Feiteira. Generation of computational components for health applications from natural language</i>	Nov 2024 - Dez 2025
Undergraduate Research Assistant Federal University of Rio Grande do Sul <i>Cauã T. D. Miranda. Automation of research instruments</i>	Sep 2024 - Present
Course Conclusion Paper Federal University of Rio Grande do Sul <i>Renato Araujo Silveira. Evolution of the Takere platform aiming for closer alignment with the language of a healthcare team</i>	Jan 2023 - Jan 2024

EXAMINATION BOARDS

Course Conclusion Paper Federal University of Rio Grande do Sul <i>Matheus Souza da Silveira. Development of a Data Pipeline for Making Anvisa Medication Information Available</i>	Jul 2026
Course Conclusion Paper Federal University of Rio Grande do Sul <i>Iago Tonello. Application of Question Answering based on LLMs and RAG for a Drug Information Center</i>	Jul 2026

COMPLEMENTARY EDUCATION

HL7® FHIR® Intermediate Course <i>Health Level Seven International</i>	2026
HL7® FHIR® Fundamentals Course <i>Health Level Seven International</i>	2026
Microfrontends with React: A Complete Developer's Guide <i>Udemy</i>	2025
Microservices with Node JS and React <i>Udemy</i>	2025
Conflicts of Interest <i>CITI Program</i>	2024
Data or Specimens Only Research <i>CITI Program</i>	2024

Google Cloud Computing Foundations ACADEMY <i>Google</i>	2022
Agile Development with Design Patterns <i>Coursera</i>	2022
React Native <i>B7Web</i>	2021
Spring Boot, Hibernate, Rest, Ionic, JWT, S3, MySql, MongoDB <i>Udemy</i>	2021
C# Object Oriented Programming + Projects <i>Udemy</i>	2021
Machine Learning and Data Science with Python from A to Z <i>Udemy</i>	2020
Object-Oriented PHP <i>B7Web</i>	2021
Object-Oriented with Java <i>Coursera</i>	2020
HTML5, CSS3, and JavaScript <i>B7Web</i>	2020

PARTICIPATION IN EVENTS

10th International Workshop on Green and Sustainable Software (GREENS'26) <i>Workshop</i>	2026
International Workshop on Agentic Engineering (AGENT 2026) <i>Workshop</i>	2026
5th International Conference on AI Engineering – Software Engineering for AI (CAIN) <i>Conference</i>	2026
13th International Conference on Mobile Software Engineering and Systems (MOBILESoft) <i>Conference</i>	2026
IEEE 38th International Symposium on Computer-Based Medical Systems (CBMS) <i>Symposium</i>	2025
Brazilian Symposium on Computing Applied to Health (SBCAS) <i>Symposium</i>	2025
Future & Innovation Health Forum <i>Meeting</i>	2025
Digital Health Forum: Data and AI <i>Meeting</i>	2025
Health Meeting <i>Fair</i>	2024
Brazilian Symposium on Software Engineering (SBES) <i>Symposium</i>	2022
Workshop on the Implications of Computing in Society (WICS) <i>Workshop</i>	2022
Brazilian Symposium on Software Engineering (SBES) <i>Symposium</i>	2021